Q1: NOV 2001 P2

4 A network of resistors, each of resistance 6.0 Ω, is constructed as shown in Fig. 4.1.

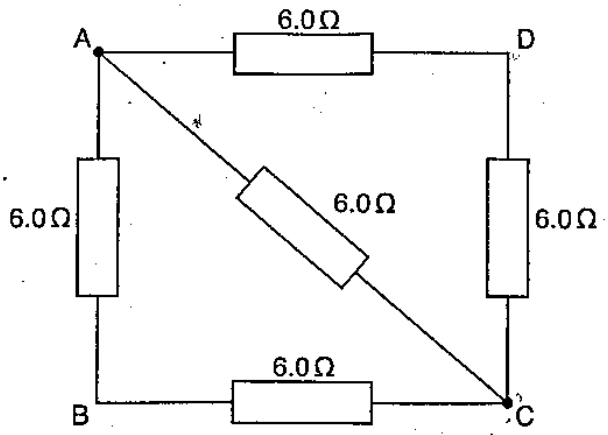


Fig. 4.1

Determine the total resistance of the network between

(a) terminals A and C,

resistance = ... Ω [2]

(b) terminals A and D.

resistance = ... Ω [3]

Q2:NOV 2004 P2

4 **(a)** State Kirchhoffs' laws.

____

____

____

____ [2]

(b) **Fig. 4.1** shows a circuit with resistors R_1 and R_2 of resistance 2 Ω and 3 Ω respectively. The circuit includes three cells of e.m.f. $E_1 = 12$ V, $E_2 = 10$ V and $E_3 = 8$ V. Each of the cells has an internal resistance of 1 Ω.

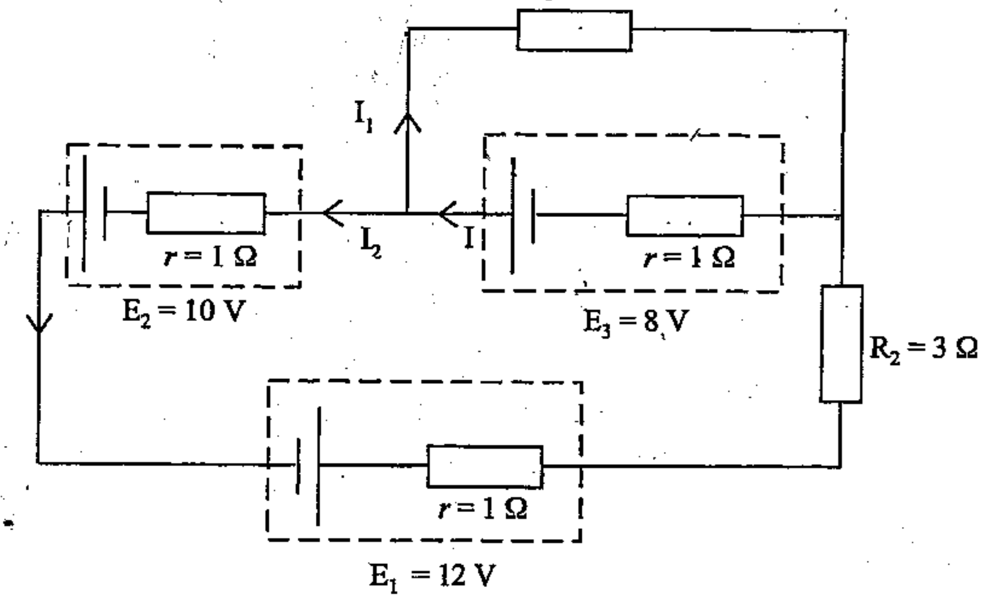


Fig. 4.1

(i) Use arrows to identify **two** loops in **Fig. 4.1.** [2]

(ii) Given that $I_1 = 1.06$ A, calculate I_2.

I_2 = ______________________________ [4]

Q3:NOV 2009 P2

5 **(a)** A potentiometer is used to measure *e.m.f.* and *p.d.* to a high accuracy.

Distinguish between *e.m.f.* and *p.d.* in terms of energy considerations.

____

____

____ [2]

(b) **Fig. 5.1** shows a potentiometer circuit for measuring the e.m.f. of a test cell X.

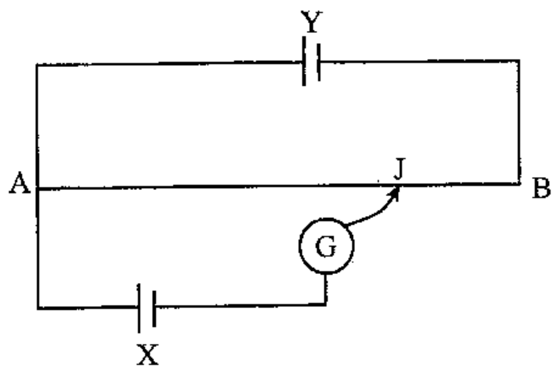


Fig. 5.1

(i) Identify cell Y and its function.

____

____ [2]

(ii) The slider J should not be moved continuously along the wire.

Explain why.

____

____ [1]

Q4:JUN 2011 P2

6 (i) State Kirchoff's **two** laws.

1. ______________________________

____

2. ______________________________

____

(ii) State the respective quantity conserved in each law.

1. ______________________________

2. ______________________________ [4]

Q5:JUN 2014 P2

6 **(a)** State and explain Kirchhoff's first and second laws.

first law:

____

explanation:______________________________

____

second law:

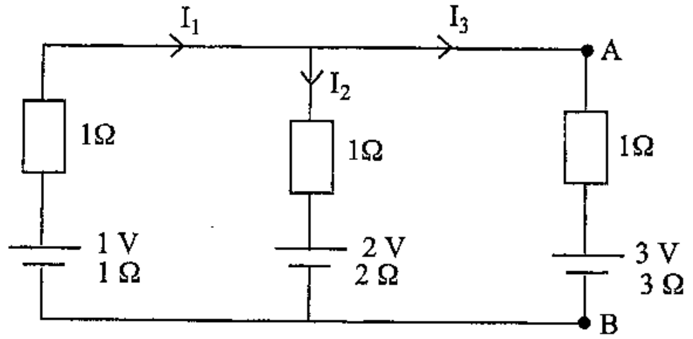
____

*explanation:*______________________________

____

[4]

(b) **Fig. 6.1** shows a simple circuit.



(i) Calculate the values of I_1, I_2 and I_3.

I_1 = ____________________

I_2 = ____________________

I_3 = ____________________

[6]

(ii) Hence determine the potential difference between points A and B.

p.d. = ________________ [1]

Q6:JUN 2017 P2

7

4 **(a)** A cell has an e.m.f. of 3V.

F
Exami
U

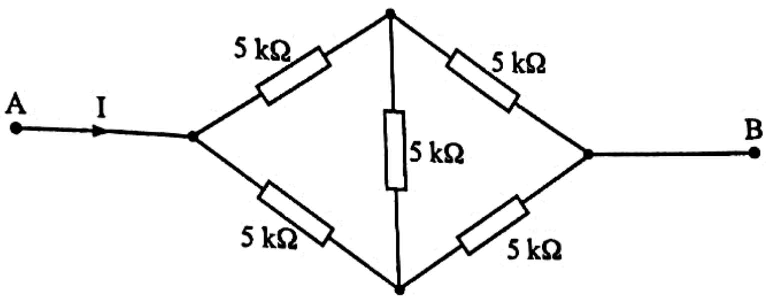
Explain the underlined phrase.

____

____

____ [1]

(b) **Fig. 4.1** shows a connection of resistors of 5 kΩ each.



The *p.d* across A and B is 3 V.

Determine the

(i) total resistance between A and B,

resistance = ____________

(ii) current, I, in the circuit.

current = ____________

[5]

Q7:NOV 2018 P2

5 The LDR in **Fig. 5.1** has a resistance of 1 KΩ in light and 1 M Ω in darkness.
The battery has negligible internal resistance.

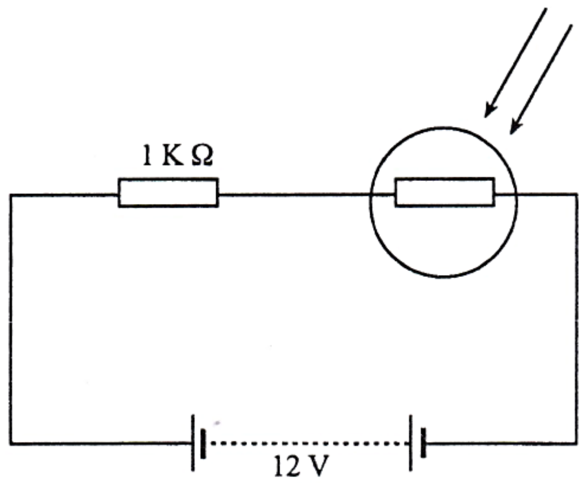


Fig. 5.1

(a) Calculate the voltage across the LDR when in darkness.

(b) A student connects a 12 V lamp of resistance 10 Ω across the LDR so that the lamp will automatically light in darkness and turn off in bright light.

State and explain whether the student will be successful or not.

____

____

____

____

____

____

Q8:2018 SPEC P2

5. When a slide wire potentiometer is used to measure the *emf* of a cell, a balance point cannot be found along the resistance wire.

Explain how a voltmeter may be used to discover the cause of the problem.

____

____

____

____ [3]

Q9:NOV 2002 P3

4 (f) A light-dependent resistor (LDR) is connected in series with a 10 kΩ resistor and a 12 V d.c. supply as shown in **Fig. 4.1**.

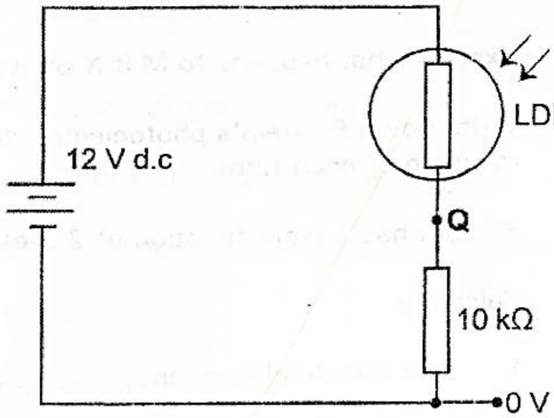


Fig. 4.1.

Find the potential at the point Q when the LDR is

(i) in the dark and has a resistance of 60 kΩ.

(ii) in bright sunlight and has a resistance of 2 kΩ.

(g) What is the resistance of the LDR if the potential at Q is 3.5 V.

(h) Two lamps **M** and **N** labelled 3 V, 0.20 A and 6 V, 0.06 A respectively, light up with normal brightness (see **Fig 4.2**).

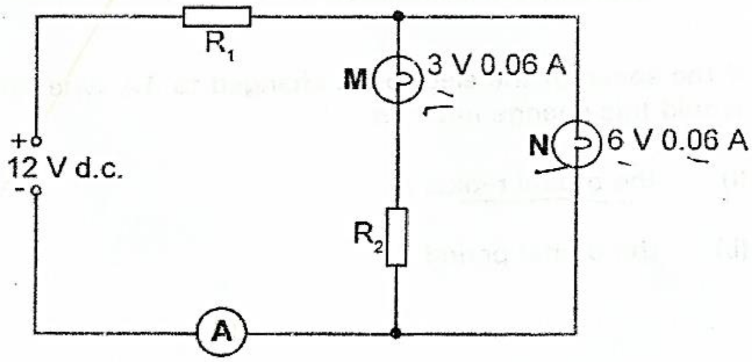


Fig. 4.2

(i) Determine the values of R_1 and R_2.

(ii) What is the ammeter reading?

(iii) Explain what happens to **M** if **N** blows out. [4]

Q10:NOV 2007 P3

4 (a) Define *potential difference* and *the volt.* [2]

(b) A voltmeter was connected to measure the potential difference across the 10.0 kΩ resistor as in **Fig. 4.1**. The potential difference was found to be 3.1 V.

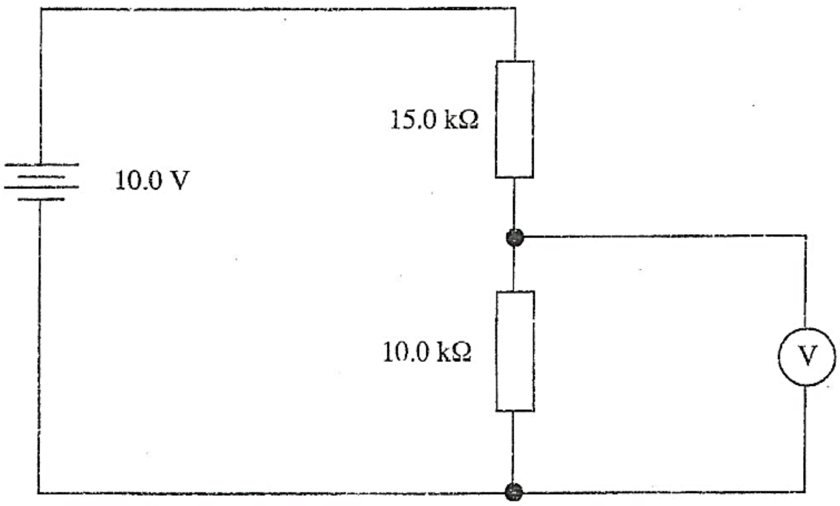


Fig. 4.1

(i) Calculate the potential difference across the 10.0 kΩ resistor before the voltmeter is connected.

(ii) Compare and comment on the two values of the potential difference.

(iii) Calculate the resistance of the voltmeter. [7]

Q11:JUN 2019 P3

3 (a) (i) Define *potential difference*.

(ii) State and explain the effect of internal resistance of a power source on its terminal potential difference.

(iii) **Fig. 3.1** shows a current of 1.5 A being driven through a circuit by a battery of emf 9.0 V and internal resistance 2.0 Ω.

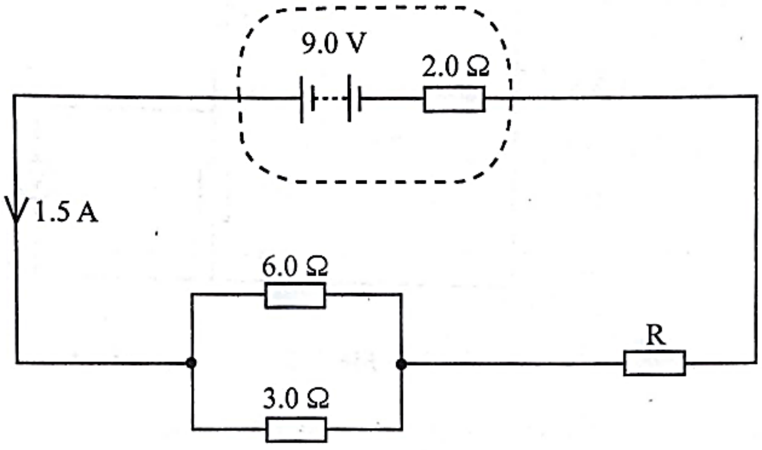


Fig. 3.1

Determine the

1. current flowing through the 6.0 Ω resistor,

2. amount of charge that passes through the 6.0 Ω resistor in 5.0 minutes,

3. value of the resistor, R.

[8]

(b) **Fig. 3.2** shows a potential divider circuit with a cell of emf 1.5 V and negligible internal resistance. The voltmeter has an infinite resistance. At 0 °C the thermistor has a resistance of 3.9 kΩ and the voltmeter reads 1.0 V.

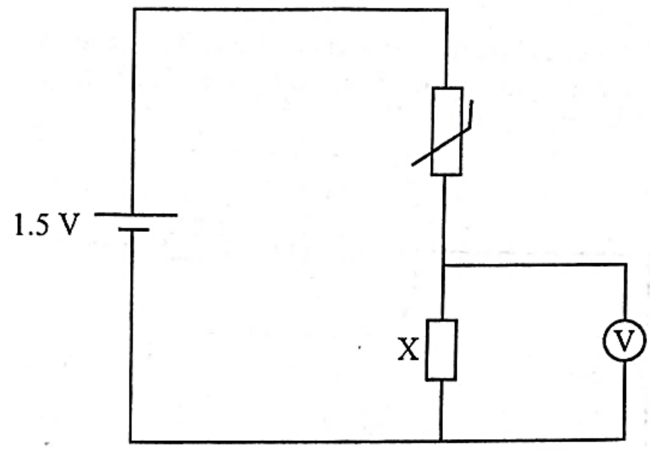


Fig. 3.2

(i) Determine the

1. resistance of the fixed resistor X,

2. voltmeter reading at 30° C when the thermistor has a resistance of 1.3 kΩ.

(ii) Suggest a practical use of this circuit.

[5]

Q12:NOV 2009 P5

2 (a) State *Kirchoff's laws.* [2]

(b) **Fig. 2.1** shows a circuit for the supply of energy to a load of 6.0 Ω.

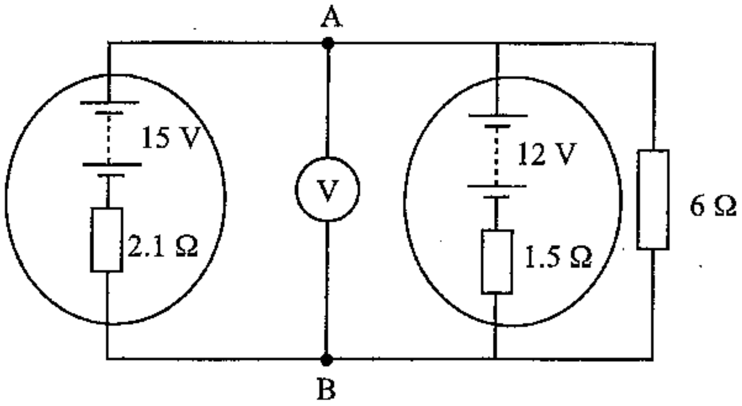


Fig. 2.1

A high resistance voltmeter is connected to the points **A** and **B**.

(i) Calculate the currents through the power supplies and the load.

(ii) Deduce the voltage recorded by the voltmeter.

(iii) Calculate the power dissipated in the load.

[10]

Q13:JUN 2009 P5

2 (a) State Kirchoff's laws and give the principle from which each is derived. [4]

(b) **Fig. 2.1** shows a circuit in which the voltmeter is used to measure p.d.

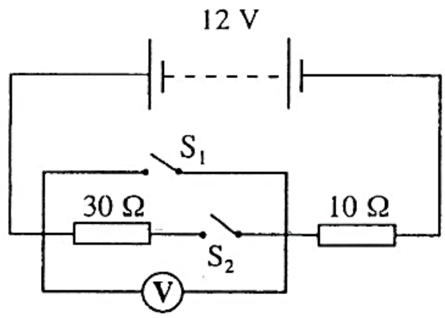


Fig. 2.1

State, giving reasons, the voltmeter reading when

(i) only S1 is closed,

(ii) only S2 is closed,

(iii) both S1 and S2 are open,

(iv) both S1 and S2 are closed.

[4]

Q14:NOV 2011 P5

3 (a) Draw a circuit symbol for a galvanometer. [1]

(b) Using a standard cell with e.m.f. of 1.0186 V, explain how you would find the e.m.f. E_1 of a torch cell using a potentiometer. [3]

(c) **Fig. 3.1** shows current flowing through a junction. **Fig. 3.2** shows a closed loop.

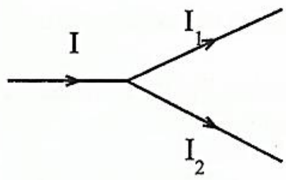


Fig. 3.1

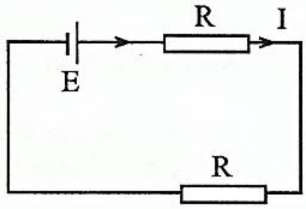


Fig. 3.2

(i) Explain why

1. $I = I_1 + I_2$,

2. $E = \Sigma IR$.

(ii) Use the concepts in (i) to find current I_1 and I_2 in **Fig. 3.3**.

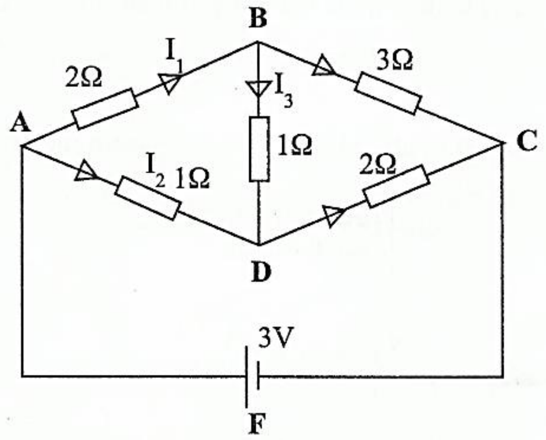


Fig. 3.3

[8]

Q15:JUN 2013 P5

1 (b) Fig. 1.1 shows a circuit with a power supply of e.m.f, E, and negligible internal resistance.

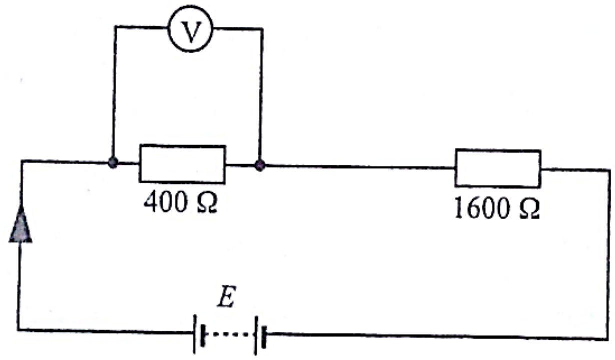


Fig. 1.1

(i) If the resistance of the voltmeter is 2 000 Ω, calculate the magnitude of E when the voltmeter gives a reading of 8 V.

(ii) Suggest **one** assumption made besides that of negligible internal resistance.

[4]

Q16:JUN 2014 P5

2 (a) (i) State Kirchoff's laws and in each case indicate the significance of the law.

(ii) Show that the combined resistance, R_T, of three resistors connected in series is given by $R_T = R_1 + R_2 + R_3$.

(iii) A battery charger is connected to a rechargeable cell as shown in **Fig. 2.1**.

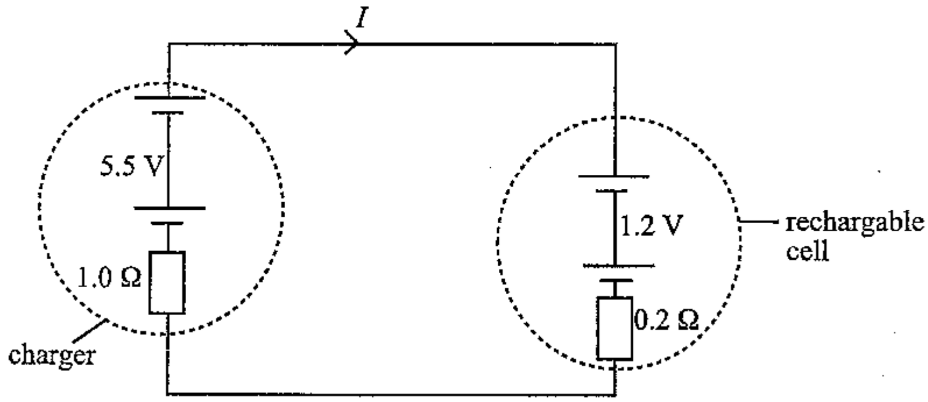


Fig. 2.1

Calculate the current through the rechargeable cell.

[8]

(b) Sketch and explain the *I-V* characteristic for a semi-conductor diode. [4]

Q17:NOV 2014 P5

2 (a) (i) Define *resistance* and *ohm*.

(ii) Calculate the electrical resistance of a wire of length 0.75 km, cross-sectional area 60 cm^2 and resistivity 1.0×10^{-7} Ωm.

[4

(b) A light dependent resistor in the potential divider circuit shown in Fig. 2.1, has a resistance which varies between 1 000 Ω in bright light and 5.00 MΩ in darkness.

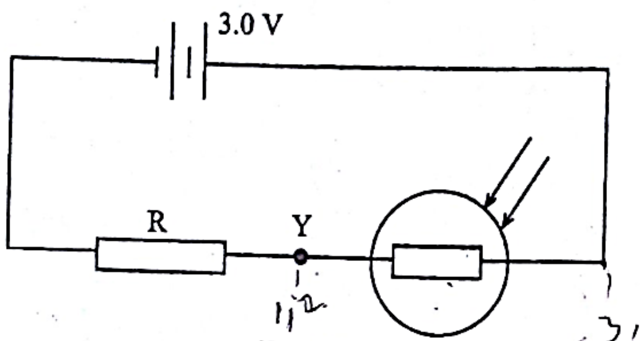


Fig. 2.1

Calculate the value of the series resistor, R, which is needed so that the potential at Y is 1.2 mV in darkness. [3]

(c) A strain gauge is securely fixed to a loaded cylinder wall under strain test and connected to a circuit as shown in **Fig. 2.2**.

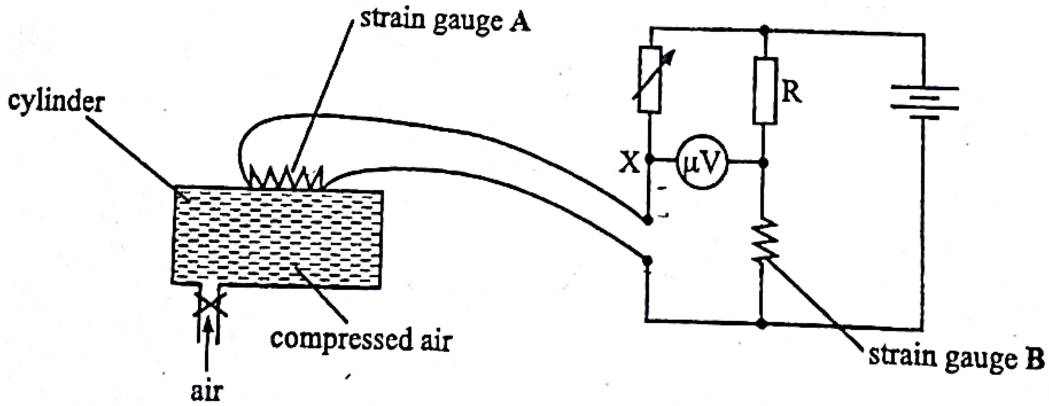


Fig. 2.2

(i) Describe and explain the changes in potential at X as the amount of compressed air is increased.

(ii) State the advantage of using two strain gauges in this arrangement.

(iii) Suggest another practical use of a similar circuit to that in **Fig. 2.2**.

9188/5 N2014 [5]